

B13A101

Diagnostic information - amplifier

Front left/right loudspeaker - woofer - electrical fault

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Open circuit: resistance > **20 ohm**
- Short circuit to ground/B+: resistance < **3 ohm**
- Short circuit between lines: resistance < **0.5 ohm**

Possible fault cause(s)

- Open circuit
- Short circuit to ground/B+ in lines
- Short circuit between lines
- Lines and/or connectors corroded
- Loudspeaker diaphragm faulty
- Internal fault in loudspeaker
- Implausible signal

Fault effect(s)

- Loudspeaker does not work
- Changed sound

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V** ⇒ **+ 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance □ **0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance < **2 ohm**

B13A401

Diagnostic information - amplifier

Rear left/right loudspeaker - mid-range - electrical fault

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Open circuit: resistance **> 20 ohm**
- Short circuit to ground/B+: resistance **< 3 ohm**
- Short circuit between lines: resistance **< 0.5 ohm**

Possible fault cause(s)

- Open circuit
- Short circuit to ground/B+ in lines
- Short circuit between lines
- Lines and/or connectors corroded
- Loudspeaker diaphragm faulty
- Internal fault in loudspeaker
- Implausible signal

Fault effect(s)

- Loudspeaker does not work
- Changed sound

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V ⇒ + 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **□ 0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance **< 2 ohm**

B13A501

Diagnostic information - amplifier

Rear left/right loudspeaker - mid-range - electrical fault

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Open circuit: resistance **> 20 ohm**
- Short circuit to ground/B+: resistance **< 3 ohm**
- Short circuit between lines: resistance **< 0.5 ohm**

Possible fault cause(s)

- Open circuit
- Short circuit to ground/B+ in lines
- Short circuit between lines
- Lines and/or connectors corroded
- Loudspeaker diaphragm faulty
- Internal fault in loudspeaker
- Implausible signal

Fault effect(s)

- Loudspeaker does not work
- Changed sound

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V ⇒ + 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **□ 0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance **< 2 ohm**

B13A801

Diagnostic information - amplifier

Front left/right loudspeaker - mid-range - electrical fault

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Open circuit: resistance **> 20 ohm**
- Short circuit to ground/B+: resistance **< 3 ohm**
- Short circuit between lines: resistance **< 0.5 ohm**

Possible fault cause(s)

- Open circuit
- Short circuit to ground/B+ in lines
- Short circuit between lines
- Lines and/or connectors corroded
- Loudspeaker diaphragm faulty
- Internal fault in loudspeaker
- Implausible signal

Fault effect(s)

- Loudspeaker does not work
- Changed sound

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V ⇒ + 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **□ 0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance **< 2 ohm**

B13A901

Diagnostic information - amplifier

Front left/right loudspeaker - mid-range - electrical fault

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Open circuit: resistance **> 20 ohm**
- Short circuit to ground/B+: resistance **< 3 ohm**
- Short circuit between lines: resistance **< 0.5 ohm**

Possible fault cause(s)

- Open circuit
- Short circuit to ground/B+ in lines
- Short circuit between lines
- Lines and/or connectors corroded
- Loudspeaker diaphragm faulty
- Internal fault in loudspeaker
- Implausible signal

Fault effect(s)

- Loudspeaker does not work
- Changed sound

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V ⇒ + 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **□ 0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance **< 2 ohm**

B13AA01

Diagnostic information - amplifier

Rear left/right loudspeaker - tweeter - electrical fault

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Open circuit: resistance > **35 ohm**
- Short circuit to ground/B+: resistance < **3 ohm**
- Short circuit between lines: resistance < **0.5 ohm**

Possible fault cause(s)

- Open circuit
- Short circuit to ground/B+ in lines
- Short circuit between lines
- Lines and/or connectors corroded
- Loudspeaker diaphragm faulty
- Internal fault in loudspeaker
- Implausible signal

Fault effect(s)

- Loudspeaker does not work
- Changed sound

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V** ⇒ **+ 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance □ **0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance < **2 ohm**

B13AB01

Diagnostic information - amplifier

Rear left/right loudspeaker - tweeter - electrical fault

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Open circuit: resistance > **35 ohm**
- Short circuit to ground/B+: resistance < **3 ohm**
- Short circuit between lines: resistance < **0.5 ohm**

Possible fault cause(s)

- Open circuit
- Short circuit to ground/B+ in lines
- Short circuit between lines
- Lines and/or connectors corroded
- Loudspeaker diaphragm faulty
- Internal fault in loudspeaker
- Implausible signal

Fault effect(s)

- Loudspeaker does not work
- Changed sound

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V** ⇒ **+ 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance □ **0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance < **2 ohm**

B13AC01

Diagnostic information - amplifier

Center loudspeaker - electrical fault

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Open circuit: resistance **> 20 ohm**
- Short circuit to ground/B+: resistance **< 3 ohm**
- Short circuit between lines: resistance **< 0.5 ohm**

Possible fault cause(s)

- Open circuit
- Short circuit to ground/B+ in lines
- Short circuit between lines
- Lines and/or connectors corroded
- Loudspeaker diaphragm faulty
- Internal fault in loudspeaker
- Implausible signal

Fault effect(s)

- Loudspeaker does not work
- Changed sound

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V ⇒ + 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **□ 0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance **< 2 ohm**

B13AD01

Diagnostic information - amplifier

Front left/right loudspeaker - tweeter - electrical fault

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Open circuit: resistance > **35 ohm**
- Short circuit to ground/B+: resistance < **3 ohm**
- Short circuit between lines: resistance < **0.5 ohm**

Possible fault cause(s)

- Open circuit
- Short circuit to ground/B+ in lines
- Short circuit between lines
- Lines and/or connectors corroded
- Loudspeaker diaphragm faulty
- Internal fault in loudspeaker
- Implausible signal

Fault effect(s)

- Loudspeaker does not work
- Changed sound

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V** ⇒ **+ 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance □ **0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance < **2 ohm**

B13AE01

Diagnostic information - amplifier

Front left/right loudspeaker - tweeter - electrical fault

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Open circuit: resistance > **35 ohm**
- Short circuit to ground/B+: resistance < **3 ohm**
- Short circuit between lines: resistance < **0.5 ohm**

Possible fault cause(s)

- Open circuit
- Short circuit to ground/B+ in lines
- Short circuit between lines
- Lines and/or connectors corroded
- Loudspeaker diaphragm faulty
- Internal fault in loudspeaker
- Implausible signal

Fault effect(s)

- Loudspeaker does not work
- Changed sound

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V ⇒ + 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance □ **0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance < **2 ohm**

B13AF01

Diagnostic information - amplifier

Subwoofer - electrical fault

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Open circuit: resistance **> 20 ohm**
- Short circuit to ground/B+: resistance **< 3 ohm**
- Short circuit between lines: resistance **< 0.5 ohm**

Possible fault cause(s)

- Open circuit
- Short circuit to ground/B+ in lines
- Short circuit between lines
- Lines and/or connectors corroded
- Subwoofer diaphragm faulty
- Internal fault in subwoofer
- Implausible signal

Fault effect(s)

- Loudspeaker does not work
- Changed sound

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V ⇒ + 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **□ 0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance **< 2 ohm**

B15E811

Diagnostic information - amplifier

Subwoofer control line - electrical fault

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Amplifier detects open circuit in control line
- Amplifier detects short circuit to ground/B+ in control line

Possible fault cause(s)

- Open circuit
- Short circuit to ground/B+ in lines
- Lines and/or connectors corroded
- Implausible signal

Test condition(s)

- Open circuit $\Rightarrow$ Subwoofer amplifier is activated
- Short circuit to ground $\Rightarrow$ Subwoofer amplifier is activated
- Short circuit to B+ $\Rightarrow$ Subwoofer amplifier is not activated

Fault effect(s)

- Subwoofer has limited functionality or does not work
- Changed sound

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V $\Rightarrow$ + 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance $\square$ **0.9 Mohm**

Open circuit:

◆ Setpoint resistance < **2 ohm**

B15E812

Diagnostic information - amplifier

Subwoofer control line - electrical fault

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Amplifier detects open circuit in control line
- Amplifier detects short circuit to ground/B+ in control line

Possible fault cause(s)

- Open circuit
- Short circuit to ground/B+ in lines
- Lines and/or connectors corroded
- Implausible signal

Test condition(s)

- Open circuit $\Rightarrow$ Subwoofer amplifier is activated
- Short circuit to ground $\Rightarrow$ Subwoofer amplifier is activated
- Short circuit to B+ $\Rightarrow$ Subwoofer amplifier is not activated

Fault effect(s)

- Subwoofer has limited functionality or does not work
- Changed sound

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V $\Rightarrow$ + 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance $\square$ **0.9 Mohm**

Open circuit:

◆ Setpoint resistance < **2 ohm**

B15E813

Diagnostic information - amplifier

Subwoofer control line - electrical fault

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Amplifier detects open circuit in control line
- Amplifier detects short circuit to ground/B+ in control line

Possible fault cause(s)

- Open circuit
- Short circuit to ground/B+ in lines
- Lines and/or connectors corroded
- Implausible signal

Test condition(s)

- Open circuit $\Rightarrow$ Subwoofer amplifier is activated
- Short circuit to ground $\Rightarrow$ Subwoofer amplifier is activated
- Short circuit to B+ $\Rightarrow$ Subwoofer amplifier is not activated

Fault effect(s)

- Subwoofer has limited functionality or does not work
- Changed sound

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V $\Rightarrow$ + 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **□ 0.9 Mohm**

Open circuit:

◆ Setpoint resistance < **2 ohm**

B1765F0

Diagnostic information - amplifier

Active noise reduction - engine speed - implausible signal

Diagnostic conditions

- Ignition on
- Engine speed > **1,000 rpm**
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier receives an implausible engine speed signal of < **500 rpm**

Possible fault cause(s)

- Signal implausible
- Line faulty
- Lines and/or connectors corroded
- Amplifier faulty

Fault effect(s)

- Active noise reduction limited or does not work

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V** ⇒ **+ 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance □ **0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance < **2 ohm**

B1853F1

Diagnostic information - amplifier

Authentication failed - signature/certificate

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Data transmission of the MOST components is encrypted using a protocol. The encryption cannot be verified by the communication partner.

Possible fault cause(s)

- The signature of the MOST control unit cannot be verified
- The communication certificate is not correct

Fault effect(s)

- No or restricted function of the relevant MOST component

B1853F5

Diagnostic information - amplifier

Authentication failed - signature/certificate

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Data transmission of the MOST components is encrypted using a protocol. The encryption cannot be verified by the communication partner.

Possible fault cause(s)

- The signature of the MOST control unit cannot be verified
- The communication certificate is not correct

Fault effect(s)

- No or restricted function of the relevant MOST component

B1853F8

Diagnostic information - amplifier

Authentication failed - signature/certificate

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Data transmission of the MOST components is encrypted using a protocol. The encryption cannot be verified by the communication partner.

Possible fault cause(s)

- The signature of the MOST control unit cannot be verified
- The communication certificate is not correct

Fault effect(s)

- No or restricted function of the relevant MOST component

B1853F9

Diagnostic information - amplifier

Authentication failed - signature/certificate

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Data transmission of the MOST components is encrypted using a protocol. The encryption cannot be verified by the communication partner.

Possible fault cause(s)

- The signature of the MOST control unit cannot be verified
- The communication certificate is not correct

Fault effect(s)

- No or restricted function of the relevant MOST component

Diagnostic information - amplifier

Amplifier - programming

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier reports incorrect programming

Possible fault cause(s)

- Programming incorrect or not carried out
- Software incompatible
- Amplifier faulty

Fault effect(s)

- Various function restrictions

Diagnostic information - amplifier

Amplifier - function restriction

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier detects internal faults

Possible fault cause(s)

- Software incompatible
- Amplifier faulty

Fault effect(s)

- Various function restrictions

Diagnostic information - amplifier

Amplifier - coding

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier reports incorrect coding

Possible fault cause(s)

- Coding incorrect or not carried out

Fault effect(s)

- Various function restrictions

Diagnostic information - amplifier

Amplifier - coding

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier reports incorrect coding

Possible fault cause(s)

- Coding incorrect or not carried out

Fault effect(s)

- Various function restrictions

Diagnostic information - amplifier

Amplifier - function restriction - temperature exceeded

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Temperature > 85 °C (> 185 °F)

Possible fault cause(s)

- Amplifier temperature too high
- Output stage temperature too high

Fault effect(s)

- Various function restrictions
- Reduced output-stage power output

Diagnostic information - amplifier

Amplifier - function restriction - temperature exceeded

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Temperature > 85 °C (> 185 °F)

Possible fault cause(s)

- Amplifier temperature too high
- Output stage temperature too high

Fault effect(s)

- Various function restrictions
- Reduced output-stage power output

Diagnostic information - amplifier

Amplifier - function restriction - temperature not reached

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Temperature < - 40 °C (- 40 °F)

Possible fault cause(s)

- Amplifier temperature too low

Fault effect(s)

- Various function restrictions

Diagnostic information - amplifier

Amplifier - coding

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier reports incorrect coding

Possible fault cause(s)

- Coding incorrect or not carried out

Fault effect(s)

- Various function restrictions

Diagnostic information - amplifier

Amplifier - coding

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier reports incorrect coding

Possible fault cause(s)

- Coding incorrect or not carried out

Fault effect(s)

- Various function restrictions

D10000

Diagnostic information - amplifier

Amplifier - component protection active/no basic setting

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Component protection active

Possible fault cause(s)

- Component protection not taught
- Component protection active

Fault effect(s)

- Function not available or available only to a limited extent

Diagnostic information - amplifier

Amplifier - component protection active/no basic setting

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Component protection active

Possible fault cause(s)

- Component protection not taught
- Component protection active

Fault effect(s)

- Function not available or available only to a limited extent

Diagnostic information - amplifier

Amplifier - component protection active/no basic setting

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Component protection active

Possible fault cause(s)

- Component protection not taught
- Component protection active

Fault effect(s)

- Function not available or available only to a limited extent

Diagnostic information - amplifier

Amplifier - component protection active/no basic setting

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Component protection active

Possible fault cause(s)

- Component protection not taught
- Component protection active

Fault effect(s)

- Function not available or available only to a limited extent

Diagnostic information - amplifier

Amplifier - component protection active/no basic setting

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Component protection active

Possible fault cause(s)

- Component protection not taught
- Component protection active

Fault effect(s)

- Function not available or available only to a limited extent

U101100

Diagnostic information - amplifier

Amplifier supply voltage below limit value

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier detects a supply voltage that is too low

Possible fault cause(s)

- Supply voltage too low

Fault effect(s)

- Amplifier has limited functionality

U101200

Diagnostic information - amplifier

Amplifier supply voltage above limit value

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier detects a supply voltage that is too high

Possible fault cause(s)

- Supply voltage too high

Fault effect(s)

- Amplifier has limited functionality

U101300

Diagnostic information - amplifier

Amplifier - coding

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier reports incorrect coding

Possible fault cause(s)

- Coding incorrect or not carried out

Fault effect(s)

- Various function restrictions

U101400

Diagnostic information - amplifier

Amplifier supply voltage below limit value

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier detects a supply voltage that is too low

Possible fault cause(s)

- Supply voltage too low

Fault effect(s)

- Amplifier has limited functionality